# Syllabus Unit-5

**Cloud Service Providers:**

EMC, EMC IT, Captiva Cloud Toolkit,

Google, Cloud Platform, Cloud Storage, Google Cloud Connect, Google Cloud Print, Google App Engine,

Amazon Web Services, Amazon Elastic Compute Cloud, Amazon Simple Storage Service, Amazon Simple Queue service,

Microsoft, Windows Azure, Microsoft Assessment and Planning Toolkit, SharePoint,

IBM, Cloud Models, IBM Smart Cloud,

SAP Labs, SAP HANA Cloud Platform, Virtualization Services Provided by SAP,

Sales force, Sales Cloud, Service Cloud:

Knowledge as a Service, Rack space, VMware, Manjra soft, Aneka Platform

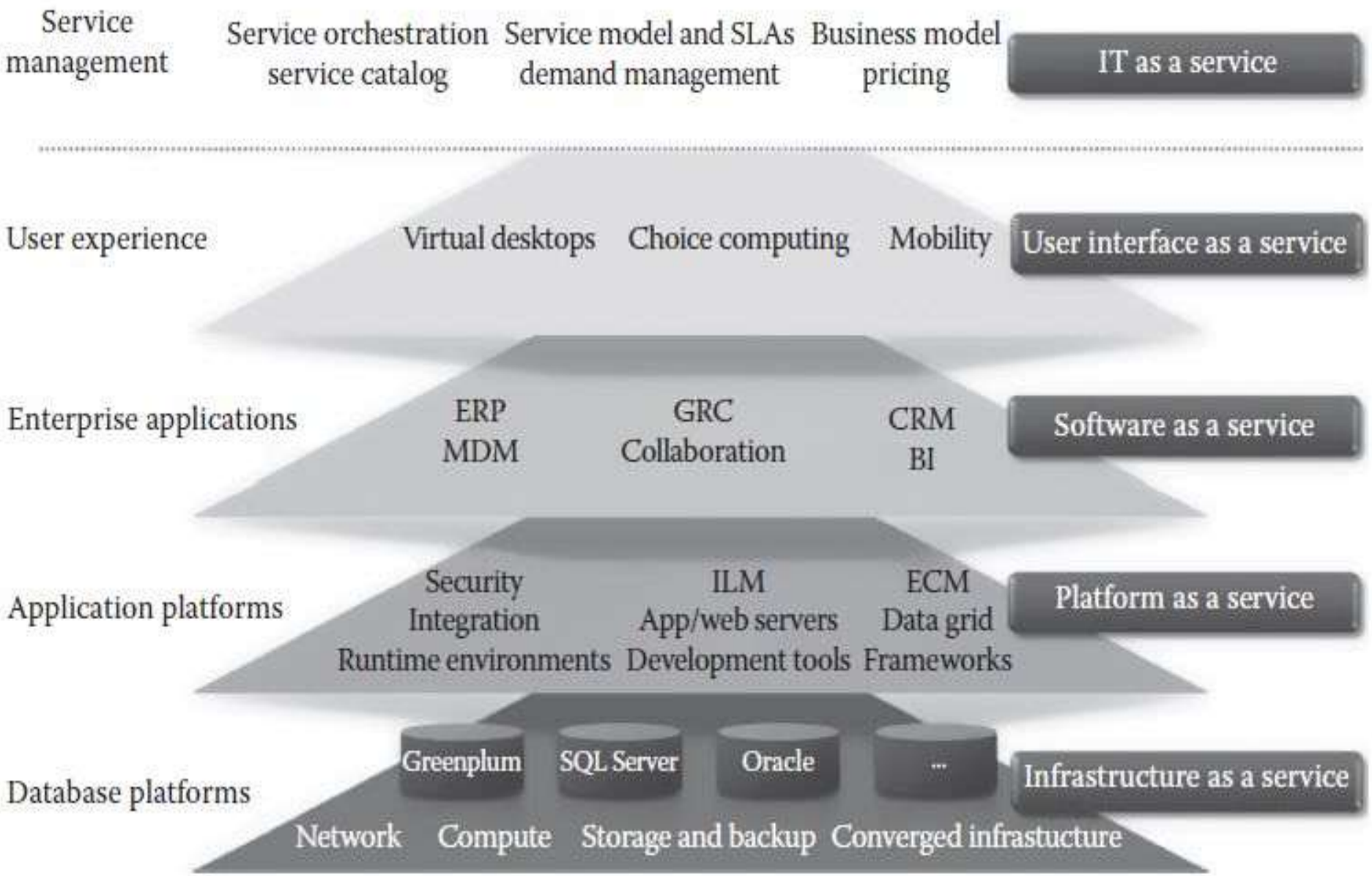
# EMC

- EMC is one of the leading global enterprises
- It require dynamic scalability and infrastructure agility to meet changing applications
- EMC choose cloud computing as the ideal solution to reduce the complexity and optimize the infrastructure
- Offering Information Technology as a Service (ITaaS)
- Reduces the energy consumption through resource sharing

# EMC IT

- Virtualization is the main concept
- Allocation of the resources on demand
- It helps to increase efficiency and resource utilization
- EMC IT provides its business process units with IaaS, PaaS, and SaaS

# Cloud services by EMC



# Cloud services by EMC

- IaaS offers EMC business units the ability to provision infrastructure components such as
  - network,
  - storage,
  - computing, and
  - operating systems individually or as integrated services

- PaaS provides the secure application and information frameworks on top of
  - application server,
  - web server,
  - database,
  - unstructured content management, and
  - security components as a service to business units

# Cloud services by EMC

- EMC IT offers
  - database platforms,
  - application platforms

- SaaS provides applications and tools

- It also offers
  - Enterprise Resource Planning (ERP) and
  - Customer Relationship Management (CRM) as a Service

- User Interface as a Service (UIaaS) provisions user and interface experience

# Captiva Cloud Toolkit

- EMC offers a tool called Captiva Cloud Toolkit

- Captiva Cloud Toolkit is a Software Development Kit (SDK)

- It helps web application developers to quickly add scanning and imaging functionality directly to their web-based business applications

- It is ideal for
  - Document capture vendors,
  - Commercial software developers,
  - and Enterprises that want to create custom web-based applications

# Captiva Cloud Toolkit Advantages

- Create a working scan-enabled web-based business application in as early as 1 week
- Time to market is shortened
- Development, testing, and support costs are greatly reduced
- Return on investment is quickly achieved
- Competitive distributed document capture market is accelerated

# Modules in Captiva Cloud Toolkit

- Scan: Scanning is the entry point to Captiva where one can import any kind of document like pdf, tiff

- MDW: Multi Directory Watch can be pointed to any folder / repository from where Captiva could import documents directly

- IE: Image enhancement is a kind of filter or repairing tool for images . The functionalities of IE are deskew, noise removal, etc

- Index: data capturing activity For example, if bank form is being processed, the A/C no. and sort code could be the indexing field

- Export: images/data are sent to various repositories like file, net, document, or data

- Multi: last process in Captiva to delete batches that have gone through all modules and exported value successfully

***Scan*:** Scanning is importing activity of documents into Captiva from a scanner. Basically, scanning happens at page level to bring images page by page into Captiva. Scanning is the entry point to Captiva where one can import any kind of document like pdf, tiff, and jpg.

***MDW***: Multi Directory Watch is another entry point to Captiva. MDW can be pointed to any folder/repository from where Captiva could import documents directly. MDW is very useful if business is getting documents in the form of a soft copy, for example, as an attached file in an e-mail. MDW also acts as a scan module except it does not inter-lock with the scanner.

***IE*:** Image enhancement is a kind of filter or repairing tool for images that are not clear. It enhances the image quality, so it could be processed easily through Captiva. One can configure IE as per business requirement and images being received. The functionalities of IE are deskew, noise removal, etc.

***Index*****:** Indexing is a data capturing activity in Captiva through which one can capture key data from various fields. For example, if bank form is being processed, the A/C no. and sort code could be the indexing field. Indexing could be added as per requirement of business. A validation field could be added to avoid unwanted data entry while indexing any document.

***Export*****:** Export is the exit point of Captiva where images/data are sent to various repositories like file, net, document, or data. The exported data are used for business requirements of various business divisions.
For example, if we are capturing the A/C no. and sort code for a bank application, this could be mapped to any department where it is needed.

***Multi*****:** Multi is the last process in Captiva to delete batches that have gone through all modules and exported value successfully. Multi could be configured as per need of business. In the case when it is required to take a backup of batches, this module could be avoided.

The previously mentioned modules are very basic modules of Captiva for indexing and exporting. But for more flexibility and automation, dispatcher is used, which is more accurate to capture data.

# Google

Google is one among the leading cloud providers that offer secure storage of user's data. It provides cloud platform, app engine, cloud print, cloud connect, and many more features that are scalable, reliable, as well as secure. Google offers many of these services for free or at a minimum cost making it user friendly.

Offer secure storage of user's data

- It provides
  - cloud platform,
  - app engine,
  - cloud print,
  - cloud connect  etc..
- Scalable, reliable, as well as secure
- Google offers many of these services for free or at a minimum cost making it user friendly

# Cloud Platform

- Google Cloud Platform enables developers to build, test, and deploy applications
- Software infrastructures such as MapReduce, BigTable, and Dremel are the innovations for industrial development
- Includes virtual machines, block storage, NoSQL datastore, and big data analytics
- Provides a range of storage services that allow easy maintenance and quick access of user's data

# Cloud Platform

- Provides easy integration of user's application within the cloud platform
- Automatically scale up to handle the most demanding workloads and scale down when traffic subsides
- The cloud platform is designed to scale like Google's own products,
- Managed services such as App Engine or Cloud Datastore
- The user has to pay only for what they uses

# Cloud Storage

- RESTful online file storage web service for storing and accessing one's data

- REST- Representational state transfer

- set of architectural constraints applied to components, connectors, and data elements within a distributed system

- The following are the few tools for Google Cloud Storage:

**Google Developers Console**- web application where one can perform simple storage management tasks

***gsutil -*** Python application that lets the user access Google Cloud Storage from the command line

# Google Cloud Connect

- It integrates cloud and the application programming interface (API) for Microsoft Office

- After installing a plug-in for the Microsoft Office suite of programs, one can save files to the cloud

- The cloud copy of the file becomes the master document that everyone uses.

- It assigns each file a unique URL that can be shared to let others view the document

- If changes are made to the document, those changes will show up for everyone else viewing it

- When multiple people make changes to the same section of a document, Cloud Connect gives chance to the user

# Google Cloud Connect

- When the user uploads a document to Google Cloud Connect, the service inserts some metadata into the file.

- Metadata is information about other information.

- In this case, the metadata identifies the file so that changes will track across all copies.

- The back end is similar to the Google File System and relies on the Google docs infrastructure.

- As the documents sync to the master file, Google Cloud Connect sends the updated data out to all downloaded copies

# Google Cloud Print

- Extends the printer's function to any device that can connect to the Internet.

- To use Google Cloud Print, the user needs to have
  - a free Google profile,
  - an app,
  - a program, or a website

  that incorporates the Google Cloud Print feature

- The print request goes through the Google servers.

- Google routes the request to the appropriate printer associated with the user's Google account

# Google Cloud Print

Assuming the respective printer is on and has

- an active Internet connection
- paper and
- ink

the print job should execute on the machine.

- The printer can be shared with other people for receiving documents through Google Cloud Print.

- Google Cloud Print is an extension built into the Google Chrome Browser, but it should be enabled explicitly

- Once enabled, the service activates a small piece of code called a connector

# Google Cloud Print

- The connector's job is to interface between the printer and the outside world

- The connector uses the user's computer printer software to send commands to the printer.

- cloud-ready printer, one can connect the printer to the Internet directly without the need for a dedicated computer

- Google allows app and website developers to incorporate Google Cloud Print into their products
- **https://www.google.com/cloudprint/learn/**

# Google App Engine

- The user run web applications on Google's infrastructure
- App Engine applications are easy to build, easy to maintain, and easy to scale as traffic and data storage needs grow
- App Engine, there are no servers to maintain: Just upload the application, and it is ready to serve users
- The app can be served from the user's own domain name (such as http:// www.example.com/) using Google Apps
- Otherwise, it can be served using a free name on the ppspot.com domain

# Google App Engine

- An application can be shared with the world or limit access to members of an organization

- Google App Engine supports apps written in several programming languages

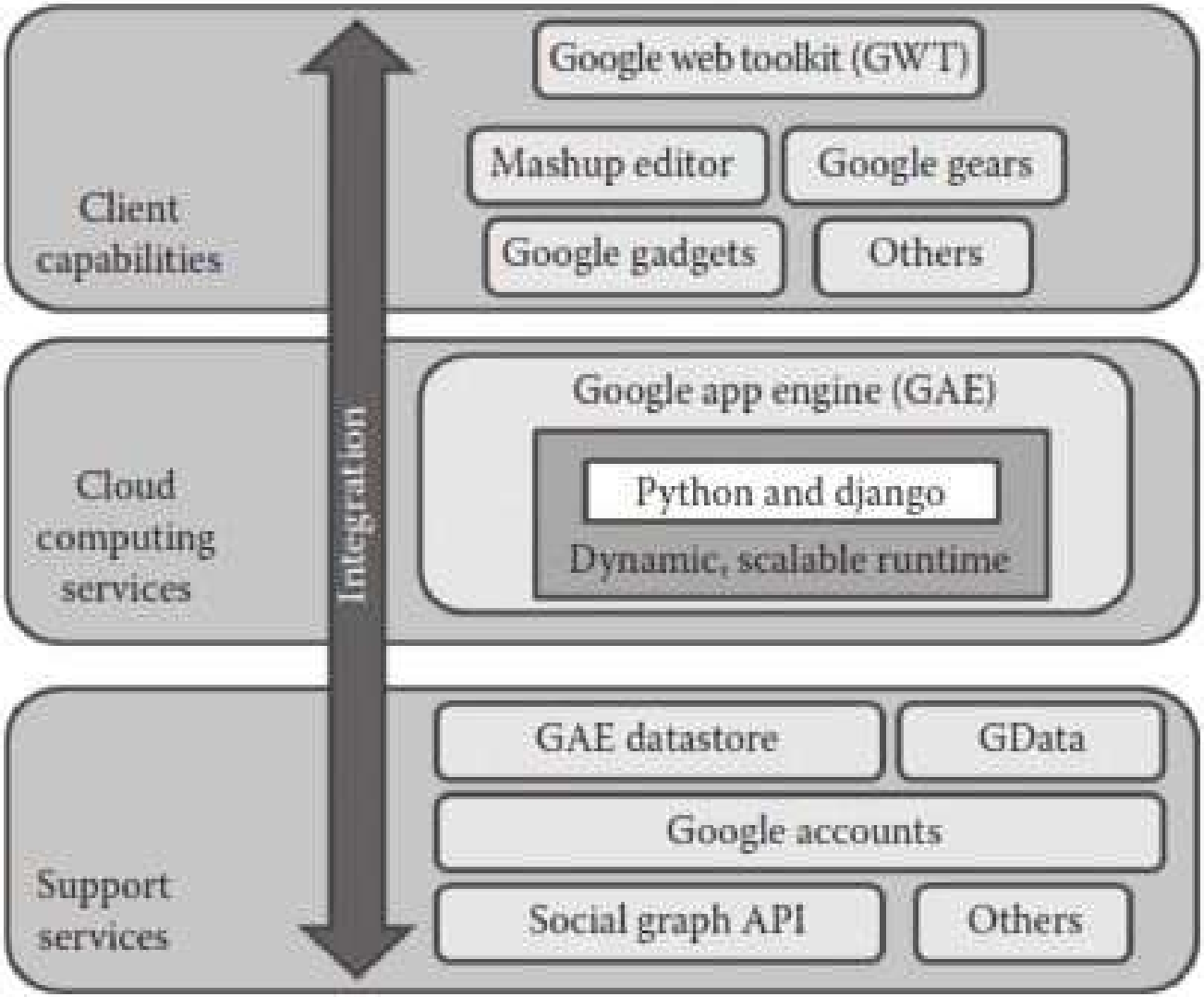
- With App Engine's Java runtime environment, one can build one's app using standard Java technologies  including JVM, servlets

- App Engine also features a Python runtime environment, which includes a fast Python interpreter and Python standard library

- App Engine also features a PHP runtime, with native support for Google Cloud SQL

- Google Cloud Storage that works just like using a local MySQL instance and doing local file writes

# Modules in Google App Engine



- Case Study
- https://cloud.google.com/appengine
- Khanacademy
- https://cloud.google.com/customers/khan-academy

# History of AWS

- 2002- AWS services launched
- 2006- Launched its cloud products
- 2012- Holds first customer event
- 2015- Reveals revenues achieved of $4.6 billion
- 2016- Surpassed $10 billon revenue target
- 2016- Release snowball and snowmobile
- 2019- Offers nearly 100 cloud services

# Amazon Web Services

**AWS** is an internet-based computing service in which large groups of remote servers are networked to allow centralized data storage, and online access to computer services or resources.

- It is a collection of remote computing services that together make up a cloud computing platform

- central and well known of these services are Amazon Elastic Compute Cloud (Amazon EC2), Amazon Simple Queue Service (Amazon SQS), Amazon Simple storage service S3

- Amazon EC2 is a computing service, whereas Amazon SQS and Amazon S3 are support services

- Provides a large computing capacity much faster and cheaper

- Amazon's data centres are located at
- Ashburn, Virginia,
- Dallas/Fort Worth,
- Los Angeles, Miami,
- Newark, New Jersey,
- Palo, Alto,
- California, Seattle,
- St. Louis, Amsterdam, Dublin,
- Frankfurt, London,
- Hong Kong, Singapore, Tokyo

- **AWS** helps companies with a wide variety of workloads **such** as game development, data processing, warehousing, achieve, development and many more. If you start an organization you will need servers that will depend on your demand. **AWS** provides a feature which allows you to choose servers according to your choice.

# AWS Services



- **There are different components of AWS, but only for key components.**
- Amazon Cluster. ...
- Storage. ...
- Databases. ...
- Management and security. ...
- Networks. ...
- Analytics. ...
- Application services. ...
- Implementation and management

# Features

- Powerful data and relationship management. ...
- Flexible schema management. ...
- Fully-managed infrastructure. ...
- Searching across objects and relationships. ...
- Built-in data encryption. ...
- Object policies.

# 5 Best Services offered by AWS are

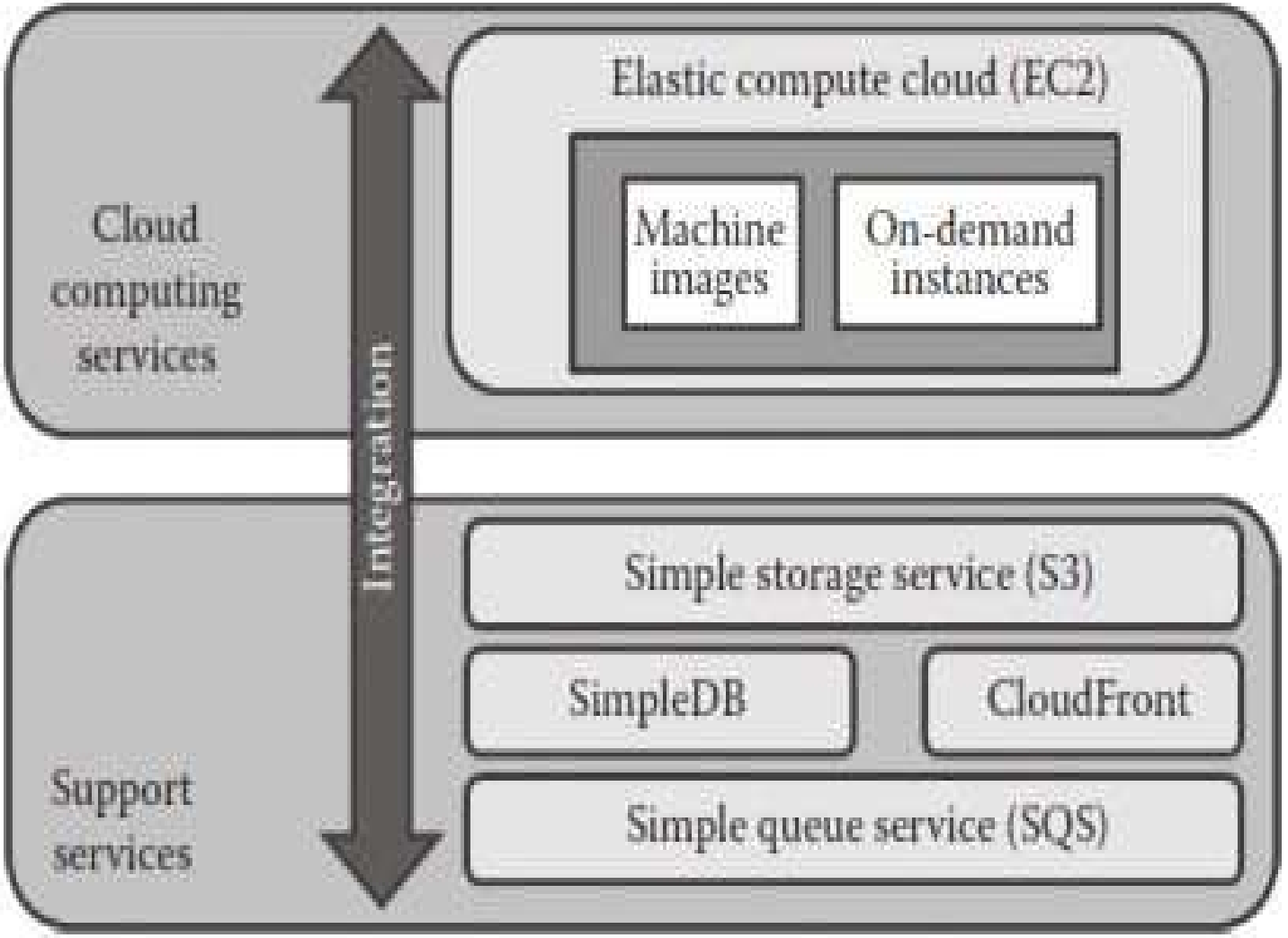
- Amazon Elastic Cloud Compute (**EC2**)
- Amazon S3 (Simple Storage **Service**)
- Amazon Virtual Private Cloud (VPC)
- Amazon CloudFront.
- Amazon Relational Database **Services** (RDS)

- **The Benefits of AWS**
- Ease of Use. ...
- Incredibly Diverse Array of Tools. ...
- Unlimited Server Capacity. ...
- Reliable Encryption & Security. ...
- Managed IT Services Are Available. ...
- **AWS** Offers Flexibility & Affordability.

# Uses of AWS

- A small manufacturing organization uses their expertise to expand their business by leaving their IT management to the AWS.
- A large enterprise spread across the globe can utilize the AWS to deliver the training to the distributed workforce.
- An architecture consulting company can use AWS to get the high-compute rendering of construction prototype.
- A media company can use the AWS to provide different types of content such as ebox or audio files to the worldwide files.

# Amazon Web Services



# Applications of AWS services

- Web site hosting
- Application hosting/SaaS hosting
- Media Sharing (Image/ Video)
- Mobile and Social Applications
- Content delivery and Media Distribution
- Storage, backup, and disaster recovery
- Development and test environments
- Academic Computing
- Search Engines
- Social Networking

# Companies using AWS

- Instagram
- Zoopla
- Smugmug
- Pinterest
- Netflix
- Dropbox

# Amazon Elastic Compute Cloud

- Amazon EC2 is an IaaS offered by AWS
- **AWS** Elastic Compute Service or **EC2** is **IaaS**(Infrastructure as a Service). This is because **Amazon** takes the responsibility of networking, storage, server and virtualization and the user is responsible for managing the Operating System, middleware, runtime, data and application.
- It provides virtual machine or instance types, operating systems, and software packages to choose from

- The user can change the capacity and characteristics of the virtual machine by using the web service interfaces

- Computing capacity is provided in the form of virtual machines or server instances by booting Amazon Machine Images (AMI)

- The primary Graphical User Interface (GUI) interface is the AWS Management Console

# Amazon Elastic Compute Cloud

- API provides programming libraries and resources for Java, PHP, Python, Ruby, Windows, and .Net

- The infrastructure is virtualized by using Xen hypervisor, and different instance types are provided as follows:

- API that supports both Simple Object Access Protocol and Query Requests
    - Standard instances—suitable for most applications
    - Micro instances—suitable for low-throughput applications
    - High-memory instances—suitable for high-throughput applications
    - High-CPU instances—suitable for compute-intensive applications
    - Cluster compute instances—suitable for HPC applications

- The instances can be obtained on demand on an hourly basis

# Amazon Elastic Compute Cloud

- provides inexpensive, low-latency network connectivity to other availability zones in the same region

- placing the instances in multiple locations enables fault tolerance and failover reliability

- Amazon EC2 instances can be monitored and controlled by the AWS Management Console and the web service API

# Amazon Elastic Compute Cloud

- AWS provides Amazon Cloud Watch, a web service that provides monitoring for AWS cloud resources

- It provides customers with visibility into resource utilization, operational performance, and overall demand patterns

- Instances are authenticated using a signature-based protocol, which uses key pairs.

- Another important feature provided is the Amazon Virtual Private Cloud (Amazon VPC)

- The existing IT infrastructure can be connected to Amazon EC2 via a virtual private network (VPN)

- Isolated computing resources are provided in Amazon VPC

# Amazon Elastic Compute Cloud

- Elastic load balancing (ELB) enables the user to automatically distribute and balance the incoming application's traffic among the running instances

- Fault tolerance and automatic scaling can be performed by configuring the ELB as per the specific needs

- Persistent storage can be enabled by using either Elastic Block Storage (EBS) or Amazon Simple Storage Service (S3)

- EBS provides a highly reliable and secure storage, and the storage volumes can be used to boot an Amazon EC2 instance

# Amazon Elastic Compute Cloud

- Amazon S3 provides a highly durable storage infrastructure designed for mission critical and primary data storage

- Storage is based on units called objects whose size can vary from one byte to five gigabytes of data

- These objects are stored in a bucket and retrieved via a unique, developer-assigned key

- It is accessible through a web service interface and provides authentication procedures to protect against unauthorized access

# Amazon Simple Storage Service

- Amazon S3 provides a simple web service interface that can be used to store and retrieve any amount of data

- It gives any developer access to the same highly scalable, reliable, secure, fast, inexpensive infrastructure

- It also takes care of other features like security, scalability, reliability, performance, and cost

- Provides a highly durable and available store for a variety of content, ranging from web applications to media files

- Amazon S3's Reduced Redundancy Storage (RRS) feature provides solution in the case of storage for data analytics

# Amazon Simple Storage Service

- Amazon S3 provides a simple web service interface that can be used to store and retrieve any amount of data

- Amazon S3 is an ideal solution for storing pharmaceutical data for analysis, financial data for computation, and images for resizing

- AWS Import/ Export feature can be used to move large amounts of data into and out of AWS with physical storage devices

- This is ideal for moving large quantities of data for periodic backups, or quickly retrieving data for disaster recovery scenarios

- Another feature offered by Amazon S3 is its Static Website Hosting
- **https://www.slideshare.net/parn09/intro-to-amazon-s3-presentation**

# Amazon Simple Queue Service

- It is a fast, reliable, scalable, fully managed message queuing service

- SQS makes it simple and cost effective to decouple the components of a cloud application

- Used to transmit any volume of data, at any level of throughput, without losing messages

- A queue is a temporary repository for messages that are waiting to be processed

- Amazon SQS offers various features like
    - allowing multiple readers and writers at the same time,
    - providing access control facilities,
    - guaranteeing high availability of sending,
    - retrieving messages due to redundant infrastructure

# APPICATIONS
# FINANCIAL SERVICES









- **Capital One** is a leading information-based technology company that is on a mission to help its customers succeed by bringing ingenuity, simplicity, and humanity to banking.
- Capital One Completes Migration from Data Centers to AWS, Becomes First US Bank to Announce Going All In on the Cloud
- We have successfully exited all of our data centers and gone all in on AWS, enabling instant provisioning of architecture and rapid innovation. We are now able to manage data at a much larger scale and unlock the power of machine learning to deliver enhanced customer experiences. On AWS, our technology teams are freed to focus on what they do best: building great software and delivering innovation to ourcustomers."

- Rob Alexander
  *Chief Information Officer, Capital One*


NFL


BBC


Discovery
CHANNEL


NETFLIX

# Media and entertainment

- **Netflix on AWS**
- Netflix is the world's leading internet television network, with more than 200 million members in more than 190 countries enjoying 125 million hours of TV shows and movies each day. Netflix uses AWS for nearly all its computing and storage needs, including databases, analytics, recommendation engines, video transcoding, and more—hundreds of functions that in total use more than 100,000 server instances on AWS.
- **Netflix Uses NICE DCV on AWS to Build VFX Studio in the Cloud for Artists Globally**
- Netflix expanded into content production in 2012 and is now one of the world's leading studios. With a culture of continual innovation, the company wanted to build a visual effects (VFX) studio in the cloud to attract top VFX and animation artists worldwide and enable seamless collaboration between global teams. Using NICE DCV and Amazon EC2 G4 Instances, Netflix built remote workstations without having to choose between responsiveness and image quality.

- **Netflix Connects with 100 Million Customers in 190 Countries Using Amazon Simple Email Service-2020**

  Before migrating to Amazon Simple Email Service (Amazon SES), Netflix maintained an in-house solution for sending email. This solution carried its own operational overhead, including running dedicated servers with email-delivery software and optimizing email-send practices for each Internet Service Provider. Netflix needed an email solution that was flexible, affordable, highly scalable, and that had global reach. Learn how Netflix uses Amazon SES to overcome these challenges and the benefits the company realized by using the service.
- **Netflix Uses Close to 1,000 Amazon Kinesis Shards in Parallel to Process Billions of Traffic Flows**

  Amazon Kinesis Streams processes multiple terabytes of log data each day

- **DISCOVERY** We migrated more than 40 sites to AWS without missing a beat. We now host all our digital media on AWS. Using the AWS Cloud gives us great capacity to expand or shrink our infrastructure as business requirements change—we now have an easy way to re-architect any of our sites.”

# Microsoft

- Azure can be described as the managed data centers that are used to build, deploy, manage the applications and provide services through a global network. The services provided by Microsoft Azure are PaaS and IaaS. Many programming languages and frameworks are supported by it.
- Azure as PaaS
- Azure as IaaS

# Microsoft

- Cloud computing provides a new way of looking at IT at Microsoft IT (MSIT)

- Cloud computing is now the preferred and default environment for new and migrated applications at Microsoft

- MSIT has developed a methodology and a set of the best practices for analyzing their current application portfolio for possible candidates to migrate to cloud computing

- This analysis enables MSIT to select the ideal cloud computing–based environment for each application

- MSIT has captured these best practices and documented them for other Microsoft customers who wish to migrate their organizations to cloud computing.

# Windows Azure

- It allow developers to easily deploy and manage application services
- It delegates the management of underlying role instances and operating system to the Windows Azure platform.
- The Migration Assessment Tool (MAT) for Windows Azure encapsulates all the information to be aware of before attempting the application migration
- By simple binary questions, the tool generates a report that outlines the amount of development effort involved to migrate the application
- The Pricing Calculator analyzes an application's potential public cloud requirements against the cost of the application's existing infrastructure
- It is available to Microsoft customers at no additional cost for installation into their data center

# Microsoft Assessment and Planning Toolkit

- MAP is an agentless, automated, multiproduct planning and assessment tool for cloud migration

- MAP provides detailed readiness assessment reports, executive proposals, and hardware and software information

- It also provides recommendations to help organizations accelerate the application migration process

- It support both private and public cloud planning assessments

- MAP analyzes server utilization data for server virtualization and also server consolidation with Hyper-V

# SharePoint

- Microsoft offers its own online collaboration tool called SharePoint

- It is a web application platform that comprises a multipurpose set of web technologies backed by a common technical infrastructure

- SharePoint has a Microsoft Office–like interface, and it is closely integrated with the Office suite

- The web tools are designed to be usable by nontechnical users

- provide intranet portals, document and file management, collaboration, social networks, extranets, websites, enterprise search, and business intelligence

- It also has system integration, process integration, and workflow automation capabilities

- https://azure.microsoft.com/en-in/case-studies/

- Spanish bank Ibercaja is helping farmers grow their businesses by providing them powerful, smart devices powered by Microsoft Azure and AI technology. By placing these IoT-enabled boxes in their fields, farmers receive data, analyzed by Azure, that helps them decide when to water their crops.
- With more than 140 years of history, Ibercaja is deeply invested in the regions where its 1,100 branch offices operate. The company's 5,500 employees have strong relationships with customers across many industry sectors, including farming. Ibercaja wanted to bring the benefits of cloud services and AI to farmers in the region: by digitally transforming the way they manage their farming businesses, the company could help them grow.The goal: to deliver detailed data about weather and moisture conditions to farmers in real time to help them make vital decisions.

- FedEx Express handles more than 6.5 million packages a day in 220 countries and territories. As the business grew and technology evolved, its customers' expectations changed, too. To streamline operations and meet customer needs with even better service, FedEx developed a package analytics solution using the machine learning operations capabilities in Microsoft Azure Machine Learning. The company reduced customer service calls, improved real-time visibility, and laid the foundation for continued innovation.
- During peak times of the pickup cycle, the company handles more than approximately 10,000 US Express packages a minute and wanted to optimize the package delivery process and predict delivery times. It determined that if it could enhance its ability to create and share analytics about all its packages, it could better serve customers and reduce customer support calls. So FedEx began to explore how to use machine learning to propel its technology capabilities forward.






- With a network of more than 174,000 students from more than 175 countries, and a mission to put a quality education within the reach of every learner, GEMS Education is one of the world's largest K-12 private education providers. Aiming to "disrupt from within" and create efficient, sustainable, scalable, and future-ready technology capabilities for the education needs of today and tomorrow, GEMS Education's

- Scaling flexibly to meet demand
- “You can imagine the early morning digital onslaught we experience, when something like 50,000 or 60,000 students suddenly sign in to the environment for attendance between seven or seven-thirty in the morning,” says Shabeer Mohammed, Head of Technology, GEMS Education. “We needed an environment powerful and flexible enough to manage that load. Sure, we could have purchased any number of servers and put them in our datacenter and then introduced some caching mechanism. But that would have meant spending an hour every morning monitoring and provisioning. Right now, Azure gives us the flexibility of using an environment which is highly adaptable—using Load Balancer or Application Gateway, for example—it can grow and shrink, depending on traffic flows, and we pay for what we use. This is very helpful to manage the morning spike—especially since system loads begin to ease by, say, eight o’clock. It made sense to use the Azure [cloud] platform to do it right.”

# AWS vs Microsoft vs Google

- https://intellipaat.com/blog/aws-vs-azure-vs-google-cloud/

# IBM

- Offering various cloud services to the consumers

- All offerings are designed for business use, marketed under the name IBM SmartCloud

- IBM cloud includes IaaS, SaaS, and PaaS offered through public, private, and hybrid cloud delivery models

- A client is designing their own virtual private cloud, deploying cloud service, or consuming cloud workload applications

- The IBM cloud framework begins with the physical hardware of the cloud

- The next layer of the IBM framework is virtualization

- IBM offers IBM Websphere application infrastructure solutions that support programming models and open standards for virtualization

# IBM

- The management layer of the IBM cloud framework includes IBM Tivoli middleware.

- Management tools provide capabilities to regulate images with automated provisioning and deprovisioning, monitor operations, and meter usage billing

- The last layer of the framework provides integrated workload tools

- Workloads for cloud computing are services or instances of code that can be executed to meet specific business needs

- **IBM offers tools for cloud-based collaboration, development and test, application development, analytics, business-to-business integration, and security.**

# Cloud Models

IBM offers five different cloud provision models:

- Private cloud, owned and operated by the customer
- Private cloud, owned by the customer but operated by IBM (or another provider)
- Private cloud, owned and operated by IBM (or another provider)
- Virtual private cloud services, based on multitenant support for individual enterprises
- Public cloud services, based on the provision of functions to individuals

# Cloud Models

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- Public cloud services, based on the provision of functions to individuals

# Cloud Models

- IBM places these offerings under three umbrellas:
  - SmartCloud Foundation,
  - SmartCloud Services,
  - and SmartCloud Solutions

- SmartCloud Foundation consists of the infrastructure, hardware, provisioning, management, integration, and security

- SmartCloud Solutions consist of a number of collaboration, analytics, and marketing SaaS applications

- IBM also offers Business Process as a Service (BPaaS)

- Infrastructure cloud services provide the consumer the provision of processing, storage, networks, and other fundamental computing

# IBM SmartCloud

- The majority of cloud users choose a hybrid cloud model,
  - with some workloads being served by internal systems,
  - some from commercial cloud providers,
  - and some from public cloud service providers

- For enterprise customers the security risk of cloud computing adoption is too high, IBM specializes in secure private cloud offerings

- For building strictly private clouds, IBM offers IBM Workload Deployer and Cloudburst as ready-to-deploy, cloud in a box–style solutions

- Cloudburst provides blade servers, middleware, and virtualization for an enterprise to build its own cloud-ready virtual machines

- Clients may choose from IBM virtualization–enabled servers, middleware, and SaaS applications

# IBM SmartCloud

- In platform cloud services, a consumer can deploy consumer-created or consumer-acquired applications

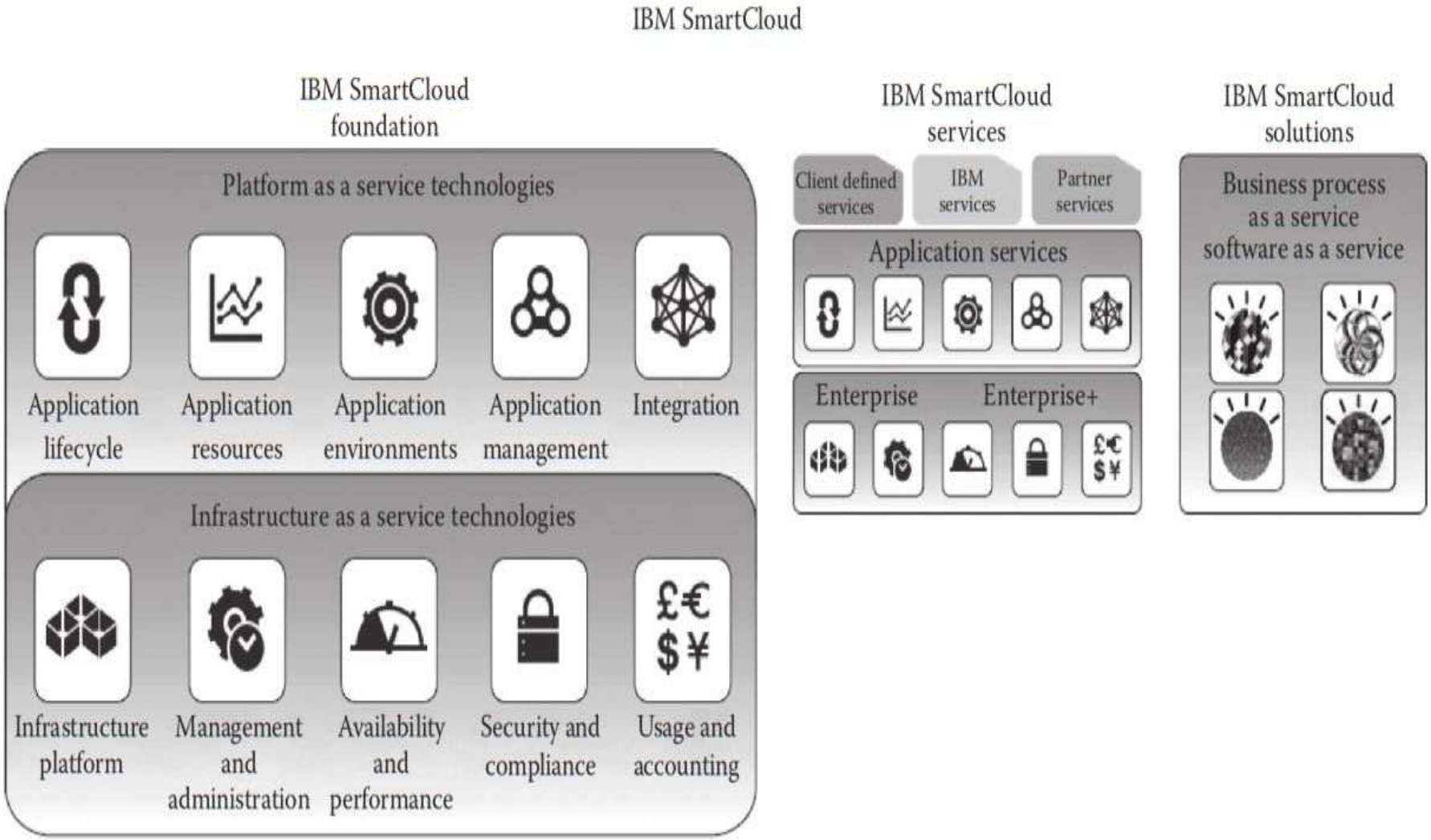
- Application cloud services allow consumers to use the provider's applications running on a cloud infrastructure

- The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based e-mail)

- Business process cloud services are any business process (horizontal or vertical) delivered through the cloud service model

- The BPaaS provider is responsible for the related business functions

# Architecture of IBM SmartCloud



- https://www.ibm.com/cloud/case-studies/
- https://www.ibm.com/products/software

# Eurobits Technologies

- To capitalize on international growth opportunities, Eurobits wanted a more flexible and scalable platform for its open banking services, with the ability to keep client data in defined jurisdictions.
- Eurobits migrated to VMware vSphere on IBM Cloud in two data centers. The company uses IBM API Connect® to industrialize API management, and IBM Cloud Kubernetes Service to containerize applications.
- Faster
- Security-rich
- Compliance
- As a pioneer in banking account information services, Eurobits helps banks work together so that their customers can easily pay from and manage their bank accounts at multiple banking providers. To support both of these activities, Eurobits provides secure APIs that enable trusted parties to securely share customer account details, and allows customers to transfer funds without logging into their own bank.

# UBank

- Established in 2008 and headquartered in Sydney, Australia, UBank is a digital-only bank. Offering services online, over the phone and through live chat, the business eliminates the costs of maintaining traditional branches and passes savings directly to customers. UBank focuses on offering simpler, smarter banking to its customers with its suite of transaction, savings and home loan products.

# American Airlines

- Customer experience is a key competitive differentiator for airlines, and increasingly depends on digital channels. How could American meet its customers' appetite for instant information and services?
- Working with IBM to migrate some of their key legacy customer-facing applications to VMware HCX on IBM Cloud, while simultaneously transforming them to a cloud-native based microservices architecture is enabling the world's largest airline to innovate faster in response to changing customer needs.

- Google Cloud Platform only offers virtual servers. IBM Cloud offers bare metal and virtual servers that can communicate with each other.
- Compared to AWS, GCP will provide you the most basic instance, containing 2 virtual CPUs and 8 GB of RAM at a 25 percent cheaper rate. ... Azure already offers pay-per-minute billing, while Google Cloud offers pay-per-second billing models which let users save way more than using AWS or Azure.

# SAP HANA

- https://www.youtube.com/watch?v=N7ym-CCgECQ

# SAP (Systems, Applications & Products)Labs

•SAP Labs makes enterprise software to manage business operations and customer relations

The company's best-known software products are:

•Enterprise resource planning application systems and management (SAP ERP),

• Its enterprise data warehouse product—SAP Business Warehouse (SAP BW),

•SAP Business Objects software,

•Sybase mobile products and in-memory computing appliance SAP HANA

# SAP HANA Cloud Platform

- open-standard, Eclipse-based, modular PaaS. In SAP HANA Cloud Platform applications are deployed via command-line tools to the cloud as web application archive (WAR) files or OSGi bundles

- OSGi bundles are normal jar components with extra manifest headers

- The applications run within the Java-based SAP HANA Cloud Platform runtime environment

- It is powered by SAP HANA and can be maintained using web-based management tools

- SAP HANA Cloud Platform lets the users quickly build and deploy business and consumer applications

# SAP HANA Cloud Platform

The main features of SAP HANA Cloud Platform are as follows:

- Enterprise platform built for developers
- Native integration with SAP and non-SAP software
- In-memory persistence
- Secure data platform
- Lightweight, modular runtime container for applications

# Virtualization Services Provided by SAP

- ERP virtualization increases a project's return on investment by maximizing hardware utilization

- The business benefits of virtualization of ERP applications are
  - shorter development cycles,
  - reduction in IT costs,
  - improved availability,
  - and energy saving

- A joint service from SAP and VMware helps in transition to a more open

- Provides flexible private cloud platform based on proven virtualization technology

# Salesforce

- Salesforce.com is a cloud computing and social enterprise SaaS provider based in San Francisco. Of its cloud platforms and applications, the company is best known for its Sales force CRM product, which is composed of Sales Cloud, Service Cloud, Marketing Cloud, Force.com, Chatter, and Work.com. In addition to its products and platforms, Salesforce.com created App Exchange, a custom application building and sharing platform. The company also has consulting, deployment, and training services.

# Salesforce

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- It is best known for its Salesforce CRM product
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# Sales Cloud

- Sales Cloud refers to the sales module in Salesforce.com

- It includes
  - Leads,
  - Accounts,
  - Contacts,
  - Contracts,
  - Opportunities,
  - Products,
  - Pricebooks,
  - Quotes,
  - Campaigns

- It includes features such as web-to-lead to support online lead capture, with autoresponse rules

- It is designed to be a start-to-end setup for the entire sales process

# Sales Cloud

- Sales Cloud manages contact information and integrates social media and real-time customer collaboration through Chatter

- The Sales Cloud gives a platform to connect with customers from complete, up-to-date account information to social insights

- It is designed use as a consumer website like Amazon and built it in the cloud to eliminate the risk and expense associated with traditional software

- Its open architecture and automatic updates eliminates the hidden costs and drawn-out implementations of traditional CRM software

- It is developing the innovate and embrace technologies like mobile, collaboration, and social intelligence

# Service Cloud: Knowledge as a Service

- Service Cloud refers to the service (as in customer service) module in Salesforce.com

- It includes Accounts, Contacts, Cases, and Solutions

- It also encompasses features such as
  - the public knowledge base,
  - web-to-case,
  - call center,
  - self-service portal,
  - customer service automation

- It delivers the world's first enterprise-grade knowledge base to run entirely on an advanced, multitenant cloud platform

- It has powerful knowledge management, without the hassle of on-premises software, is provided

# Service Cloud: Knowledge as a Service

- It has to offer all the tools one needs to run the entire service operation
- Here one can continually create, review, deliver, analyze, and improve the knowledge
- User's knowledge is available wherever other customers need it
- Agents have the right answers at their fingertips to communicate over the phone, send out through an e-mail, or share via a chat client
- The same knowledge base serves up answers to the service website is a part of company's public site
- The Service Cloud gives the tools that are needed to manage knowledge at enterprise scale
- Here user will benefit no matter what size or how complex the business is

# Rackspace

- Offering IaaS to clients, it has been used by a large number of enterprises
- It offers three cloud computing solutions—Cloud Servers, Cloud Files, and Cloud Sites
- It provide computational power on demand in minutes
- Cloud Sites are for robust and scalable web hosting
- Cloud Files are for elastic online file storage and content delivery
- Here computing capacity is provided as virtual machines that run in the Cloud Servers systems
- The virtual machine instances are configured with different amounts of capacities

# Rackspace

- The instances come in different flavors and images
- A flavor is an available hardware configuration for a server
- Each flavor has a unique combination of disk space, memory capacity, and priority for CPU time
- A varied set of instances are available for the user to choose from
- These virtual machines are instantiated using images
- An image is a collection of files used to create or rebuild a server
- A variety of prebuilt operating system images are provided by Rackspace Cloud

# Rackspace

- These images can be customized to the user's choice to create custom images
- The virtual machines that are generated come in different sizes and measured based on the amount of physical memory reserved.
- The physical memory can vary from 256 MB to 15.5 GB
- In the event of availability of extra CPU power, Rackspace Cloud claims to provide extra processing power to the running workloads, free of cost
- Backup schedules can be created to define when to create server images.
- Enables the user to continue work in the event of failures by using the backup images

# Rackspace

- Custom images are helpful in creating backup schedules.
- A type of images, referred to as gold server images, can be produced if the servers of that configuration are to be instantiated frequently
- Cloud Servers can be run through the Rackspace Cloud Control Panel (GUI) or programmatically via the Cloud Server API using a RESTful interface
- The control panel provides billing and reporting functions
- It provides access to support materials including developer resources, a knowledge base, forums, and live chat.
- The Cloud Servers API was open sourced under the Creative Commons Attribution 3.0 license

# Rackspace

- Language bindings via high-level languages like C++, Java, Python, or Ruby
- The virtual machine instances are authenticated in the API by a token-based protocol that uses the HTTP x-Header. Private/public keys
- This process is automated and initiated from either the Rackspace Cloud Control Panel or the Cloud Server API.
- The amount to scale is specified; the Cloud Server is momentarily taken offline; the RAM, disk space, and CPU allotment are adjusted; and the server is restarted
- A Cloud Server can be made to act as a load balancer using simple readily available packages
- Cloud Servers are provided persistent storage through RAID10 disk storage

# VMware

- VMware, a leader in virtualization technology,
- VMware is currently providing a range of products for the development of private and public clouds
- It leveraging the services VMware vCloud Director, VMware vCloud Datacenter Services, VMware vSphere, and VMware vShield to name a few.
- Private clouds enable the better usage and management of internal IT infrastructure than the traditional methods
- Greater operational efficiency, secure, fault-tolerant, well-managed computing environments can be modeled and operated.
- VMware's private cloud offering provides greater standardization, rapid provisioning, and self-service for all applications

# VMware

- VMware, a leader in virtualization technology,
- Private clouds can be created by using the VMware vSphere and VMware vCloud Director
- VMware vSphere is a robust virtualization platform used to transform IT infrastructures into virtual storage, compute, and network resources
- VMware vSphere provides services at both the infrastructure and application levels
- At the infrastructure level, it provides options to perform efficient operation and management of the compute, storage, and network resources
- At the application level, service-level controls are provided for the applications running on the underlying infrastructures, leading to available, secure, and scalable applications

# VMware

- The VMware vCloud Director, coupled with VMware vSphere, is a software solution that enables enterprises to build secure, multitenant private clouds

- VMware vCloud Director abstracts the virtual computing environment from the underlying resources and provides a multitenant architecture

- VMware vShield technologies are used to provide security to these environments by using services like perimeter protection, port-level firewall, NAT and DHCP services, site-to-site VPN, network isolation, and web load balancing.

- The VMware vCloud Director allows users to catalog infrastructure and application services of the desired configurations and deploy and consume them as needed

- The vCloud API is an open, REST-based API that provides scripted access, complying with the open virtualization format (OVF).

# VMware

- The API can be used along with VMware vCenter Orchestrator to automate and orchestrate operational processes like routine tasks, activities, and workflows

- vCloud Datacenter Services provides a scalable environment, where internal resources are augmented with the external resources

- vCloud Express is an IaaS offering delivered by leading VMware service provider partners

- The VMware vCloud Express providers are Virtacore vCloud Express, Hosting.com, Melbourne IT, and Terremark's vCloud Express.

- Instance types, load balancing, storage options, and pricing vary between service providers.

# Manjrasoft

- Manjrasoft is one of the nonmajor providers of cloud services
- It has come up with a platform called Aneka that provides a set of services that help the development of applications
- Manjrasoft develops marketoriented cloud computing platforms that allow one to build, accelerate, and manage the applications
- It helps ultimately saving one's time and money, leading to enhanced business productivity and profit.

# Aneka Platform

- A configurable and flexible execution platform (container) enabling pluggable services and security implementations. Multiple authentication/ authorization mechanisms such as role-based security and Windows domain–based authentication are considered for this purpose

- Multiple persistence options including Relational Database Management System (RDBMS), Structured Query Language (SQL) Express, MySQL, and flat files

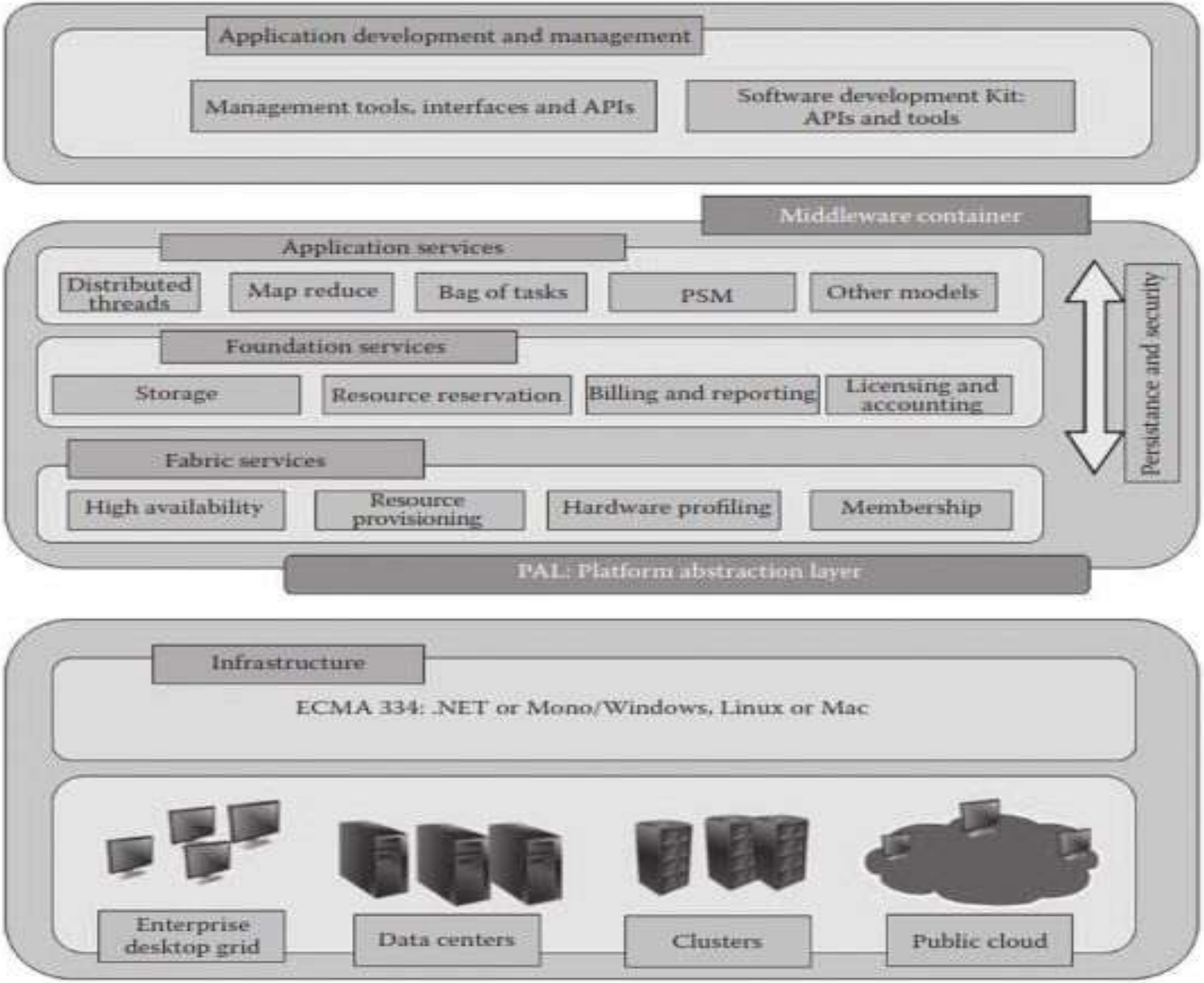
- Software development kit (SDK) supporting multiple programming models including object-oriented thread model, task model for legacy applications, and MapReduce model for data-intensive applications.

- Custom tools such as Design Explorer for parameter sweep studies

- Easy to use management tool for SLA and Quality of Service (QoS) negotiation and dynamic resource allocation

- Supports deployment of applications on private or public clouds in addition to their seamless integration

# Aneka Platform



# Aneka Platform

- Aneka allows servers and desktop PCs to be linked together to form a very powerful computing infrastructure

- This allows companies to become energy efficient and save money without investing in a number of computers to run their complex applications

- Each Aneka node consists of a configurable container that includes information and indexing, scheduling, execution, and storage services.

- Aneka supports multiple programming models, security, persistence, and communications protocols

# Summary

Tools and Services Offered by Companies

| Company Name | Tools/Services |
|---|---|
| EMC | Captiva Cloud toolkit |
| Google | Google App Engine, Google Docs, Google Cloud Connect Google Cloud Print |
| Amazon | Amazon EC2, Amazon S3, Amazon SQS |
| Microsoft | Microsoft Assessment and Planning Toolkit, Windows Azure Sharepoint |
| IBM | IBM Smart Cloud |
| Salesforce | Sales Cloud, Service Cloud |
| SAP LABS | SAP HANA Cloud |
| VMware | vCloud |
| Manjrasoft | Aneka Platform |
| Red Hat | OpenShift Enterprise, OpenShift Origin |
| Gigaspaces | Cloudify |

# Summary

Details of Cloud Service Providers

| Provider Name | Service Model | Deployment Model | Server Operating System |
|---|---|---|---|
| Amazon Web Services | IaaS | Public | Widows, Linux |
| Google App Engine | PaaS | Public | Windows |
| Windows Azure | IaaS | Public | Widows, Linux |
| IBM Cloud | IaaS | Private, hybrid | Widows, Linux |
| Salesforce Platform | PaaS | Public | Widows, Linux |
| Rackspace | IaaS | Public, private, hybrid | Widows, Linux |
| SAP HANA Cloud | PaaS | Public | Linux |

## Review Questions

1. What do you mean by cloud service provider? Which are the major cloud service providers?
2. List the tools/services provided by Microsoft and explain them in brief.
3. What is Google Cloud Print? What are its advantages?
4. Explain SAP HANA Cloud in brief.
5. What are the services offered by EMC IT? Explain.
6. Explain the services provided by IBM SmartCloud.
7. What are the support services offered by Amazon Web Services? Explain.
8. What do you mean by *Knowledge as a Service*? Which company provides this service? Explain.
9. Explain the features of Aneka.
10. What is vCloud? Explain in brief.